Midpoint & Distance Formula — Notes

Watch the video and fill in each blank. Pause when you see “Your turn.”

Name: _____ Date: _____

☞ *Video: The trick*

Key Idea

Midpoint = **M**_____ **D**istance = **D**_____

☞ *Video: Midpoint: simple case first*

Key Idea



The middle of -3 and 3 is _____.

Add and halve: $\dfrac{-3+3}{2} = 0$ ✓

So the midpoint _____ — it is the _____.

☞ *Video: Midpoint formula*

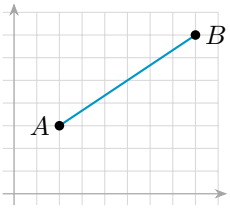
Key Idea

$M =$ _____

Example: $A(2,3)$, $B(8,7)$

x: $\dfrac{2+8}{2} =$ _____ y: $\dfrac{3+7}{2} =$ _____

$M =$ _____



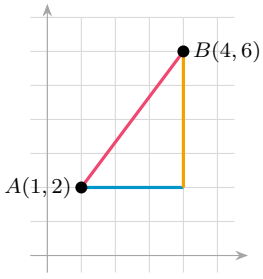
☞ *Video: Distance: simple case first*

Key Idea

From $(2,0)$ to $(5,0)$: count the squares: _____. That is $5-2$ — a _____.

☞ *Video: Distance = Pythagoras on a grid*

Key Idea



Find AB: across _____, up _____.

$d = \sqrt{3^2+4^2} = \sqrt{25} =$ _____

$d =$ _____

The distance formula is just the _____.

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☞ *Video: Two common mistakes*

Common Mistake

Square _____ difference before adding: $\sqrt{3^2+4^2}=5$, not $3+4=$ _____.

Testing add vs. subtract? Don't use 0 and 2 (both give 1). Use 2 and 4: the middle is _____ $=\dfrac{2+4}{2}$.

☞ *Video: Your turn*

Your Turn

$P(-1,4)$ and $Q(5,-4)$. Find the midpoint of $\overline{PQ}$ and the length PQ.

$M=\left(\dfrac{-1+5}{2},\ \dfrac{4+(-4)}{2}\right)=$ _____

$PQ=\sqrt{(5-(-1))^2+(-4-4)^2}=\sqrt{6^2+8^2}=\sqrt{100}=$ _____

Notice

Remember: **Midpoint = Mean**: add the coordinates, divide by 2. **Distance = Difference**: subtract, square, add, square root.

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